

Keywords: Oxygen, carbon dioxide, alveoli, lungs, red blood cells, trachea, inhale, exhale

Starter: Complete these quiz questions about last lesson

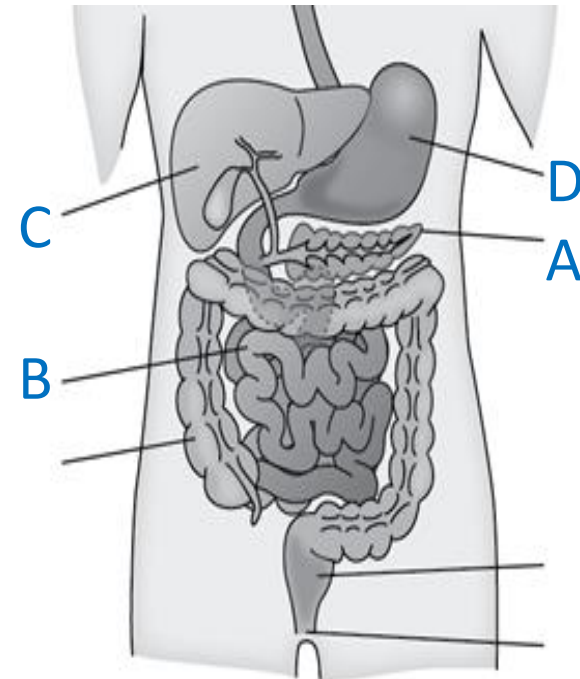
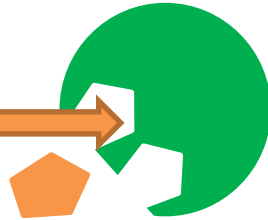
TIME FOR A

QUIZ!



Quiz: Blood

1. Name the component of blood that carries oxygen
a) platelets b) white blood cells c) red blood cells d) plasma
2. Name the component that carries dissolved substances
3. Name the component that fights disease
4. Which is not a cell that fights disease a) lymphocyte b) Terocyte
5. Which vessel carries oxygenated blood away from the heart at high pressure a) vein b) artery c) capillary
6. Which vessel has valves
7. Which vessel is only one cell thick
8. Which two vessels have muscle
9. What is the arrow pointing to on the enzyme
A _____ site
10. Name organ D and C



Breathing and Gas Exchange

08/06/2026

Keywords: Oxygen, carbon dioxide, alveoli, lungs, red blood cells, trachea, inhale, exhale

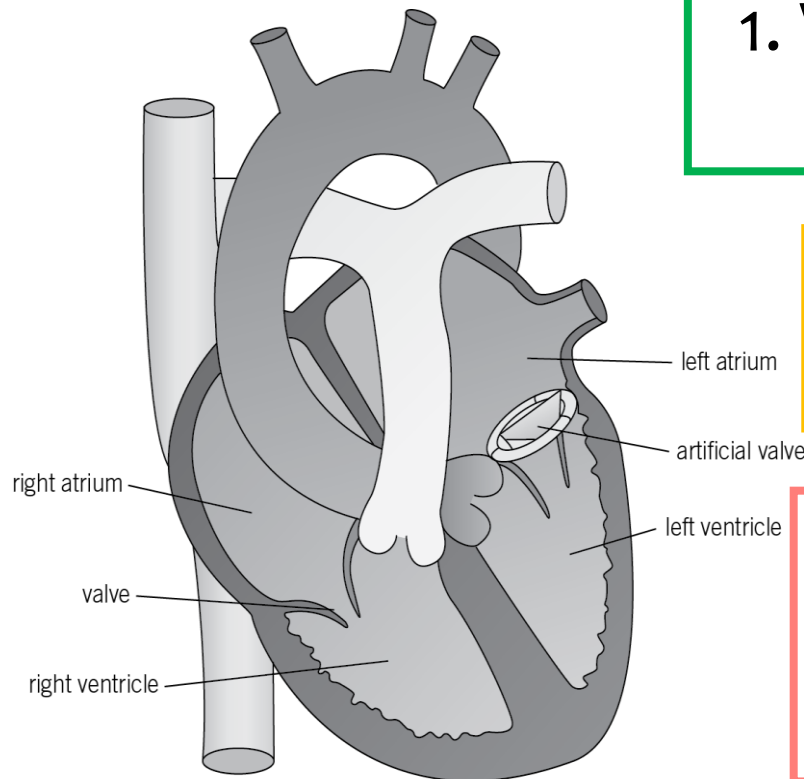
Starter: Answer these three review questions below.

Ext: What is a concentration gradient?

1. Where is the pacemaker located?
[1 mark]

2. What two types of valve are used to replace leaky valves? [2]

3. Explain when an artificial heart would be used and why. What is a disadvantage of using one? [4]



Review

1. Where is the pacemaker located? [1 mark]

Right atrium

2. What two types of valve are used to replace leaky valves? [2]

Biological tissue valve
Mechanical valve

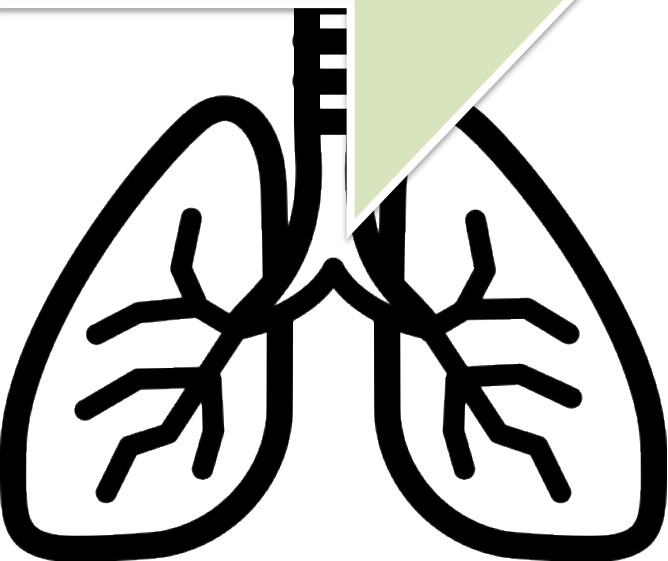
3. Explain when an artificial heart would be used and why. What is a disadvantage of using one? [3]

It is given to those who are waiting for a heart transplant. [1] It assists their damaged heart to give it rest. [1] Artificial hearts are not permanent solutions [1] and can cause blood clots. [1]

LOb: Understand the respiratory system and how it is involved in gas exchange.

Keywords:

Oxygen, carbon dioxide, alveoli, lungs, trachea, inhale, exhale



Learning Outcomes:

- Recall what gas exchange is and why we continually breathe
- Describe the structure of the gas exchange system
- Explain how parts of the gas exchange system are adapted to their function

Did you know... the lungs are not the same size. The right lung is larger than the left lung to help the heart fit into the rib cage.

When we breathe we **take in oxygen** and **give out carbon dioxide**. This is called **gas exchange**.

Exchanging means to **replace** one thing with another.

OXYGEN gas
goes in to the
lungs

As a percentage
how much
oxygen is in the
air?

10% 21% 28%?

21%

**CARBON
DIOXIDE** gas
comes out

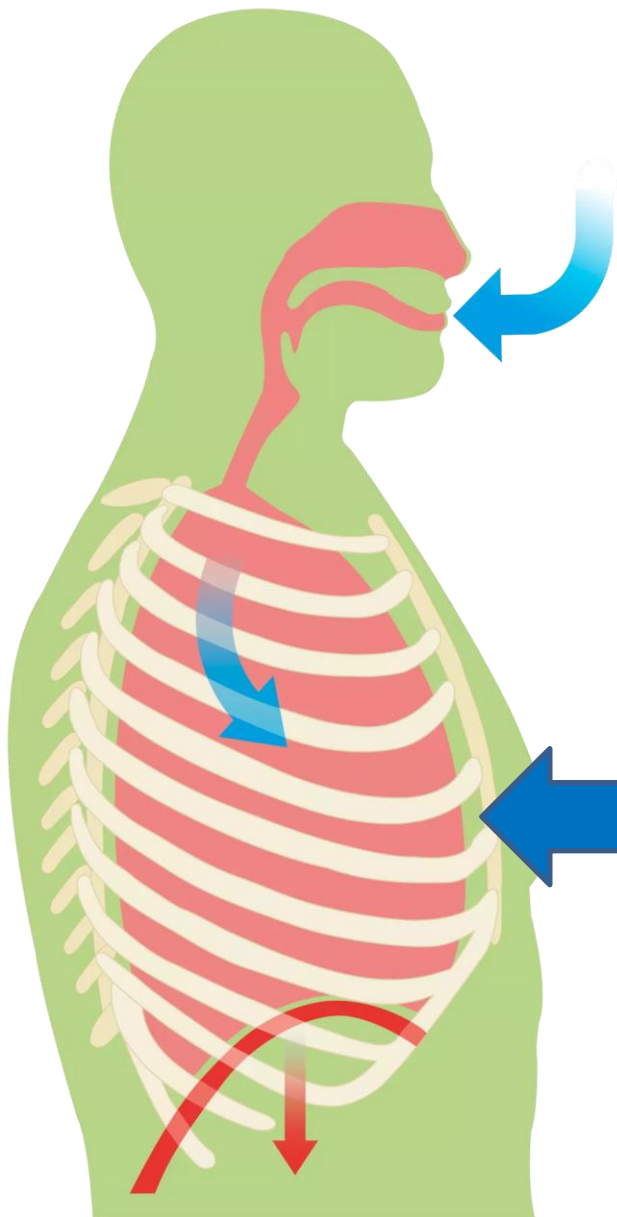


The Lungs

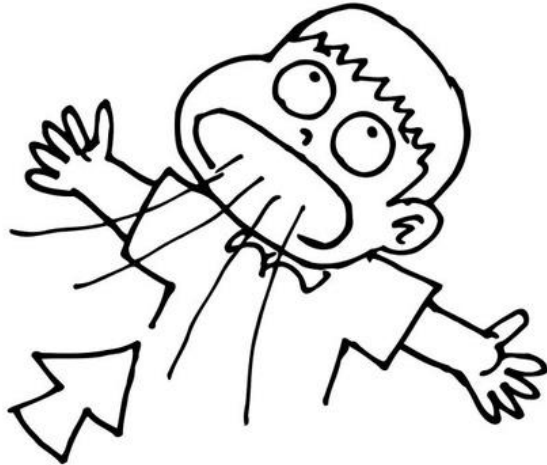
Gas exchange happens at the 'end of lungs' in air sacs called alveoli which are elastic, like a rubber band on a sling shot.



Lungs are delicate so they are protected by the hard and strong **rib cage**.



The reason we breathe regularly is to keep the **concentration of gradient** high for the gas exchange gases.



A **high concentration gradient** means O_2 and CO_2 can continually diffuse in and out of the bloodstream.

Steeper gradient = faster diffusion

Plenary: Chose a number to reveal a question related to what we have learnt.

1
How
air?

2
what
energy?

3
What
happens?

4
Why do we
continually
breathe?

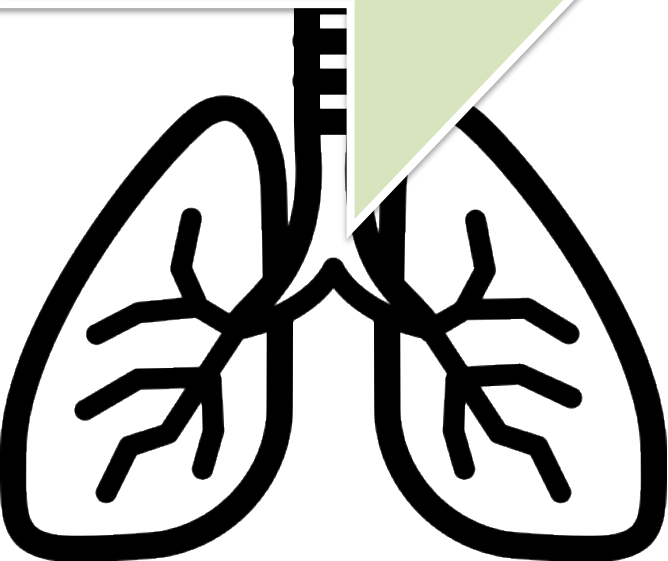
5
What is
a gradient?

6
What is
involved
with
exchange?

LOb: Understand the respiratory system and how it is involved in gas exchange.

Keywords:

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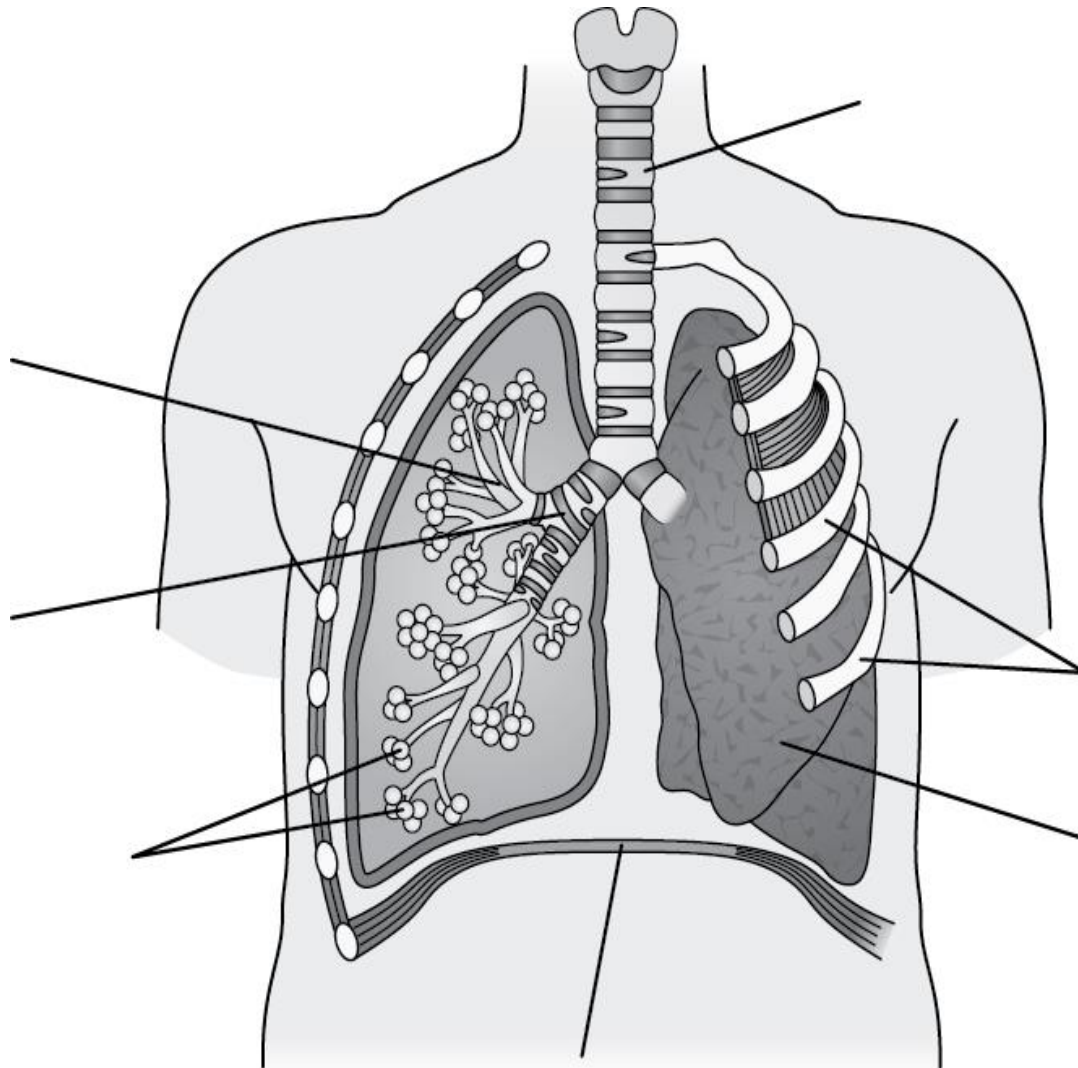


Learning Outcomes:

- ✓ Recall what gas exchange is and why we continually breathe
- Describe the structure of the gas exchange system
- Explain how parts of the gas exchange system are adapted to their function

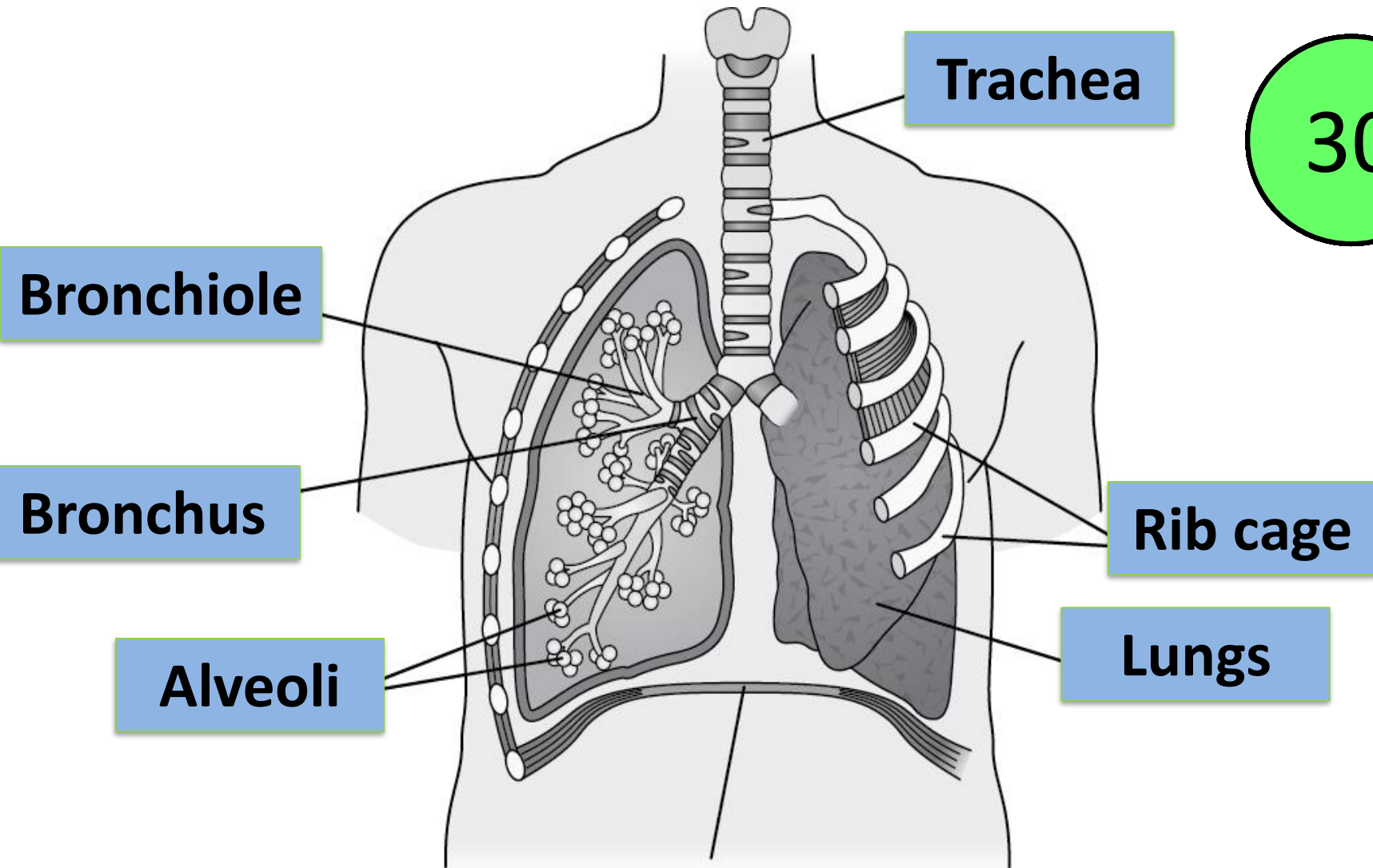
Did you know... the lungs are not the same size. The right lung is larger than the left lung to help the heart fit into the rib cage.

Here is a diagram of the respiratory system. It shows the organs and tissues involved with gas exchange.



You have thirty seconds to remember the parts.

30



**Diaphragm- Muscle under the lungs.
Pulls the air into the lungs**

Put them where they go onto the diagram.

Bronchus

Lungs

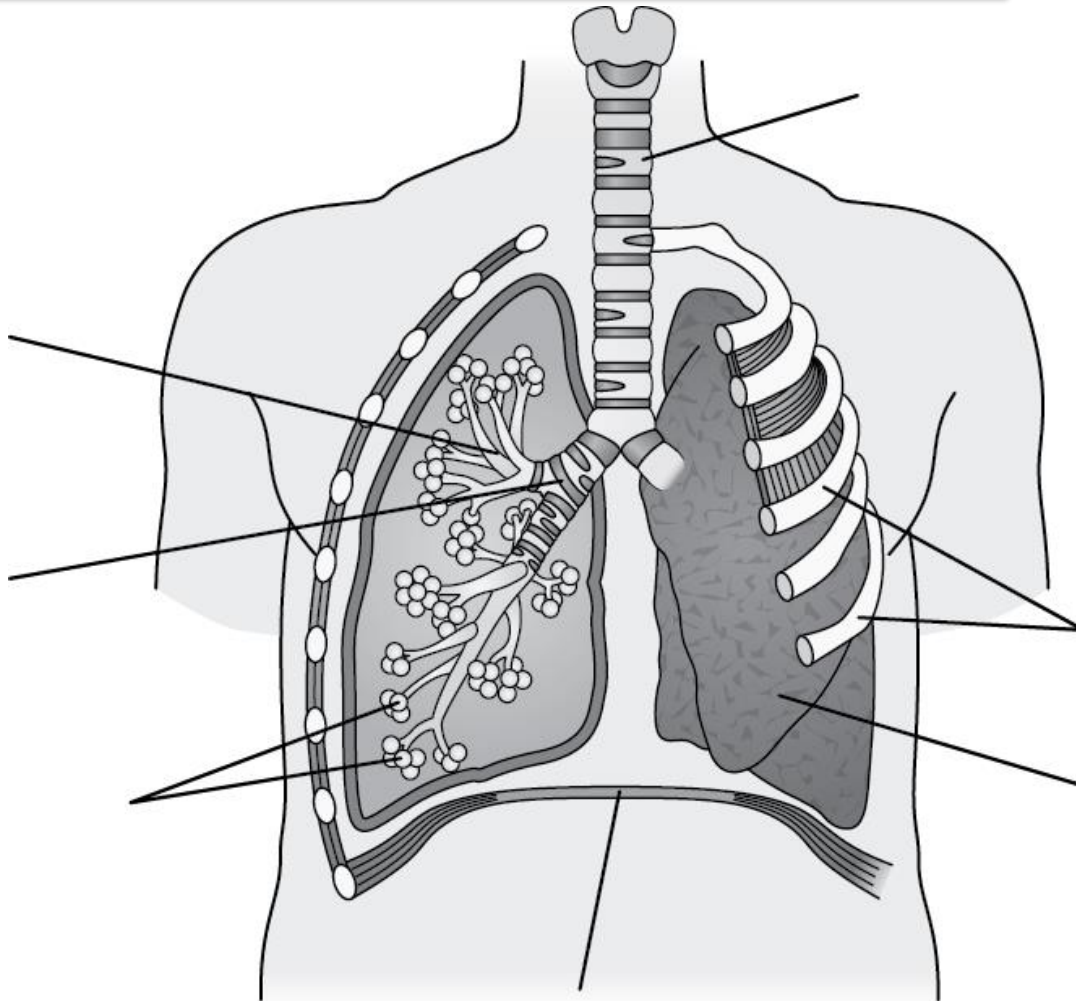
Trachea

Bronchiole

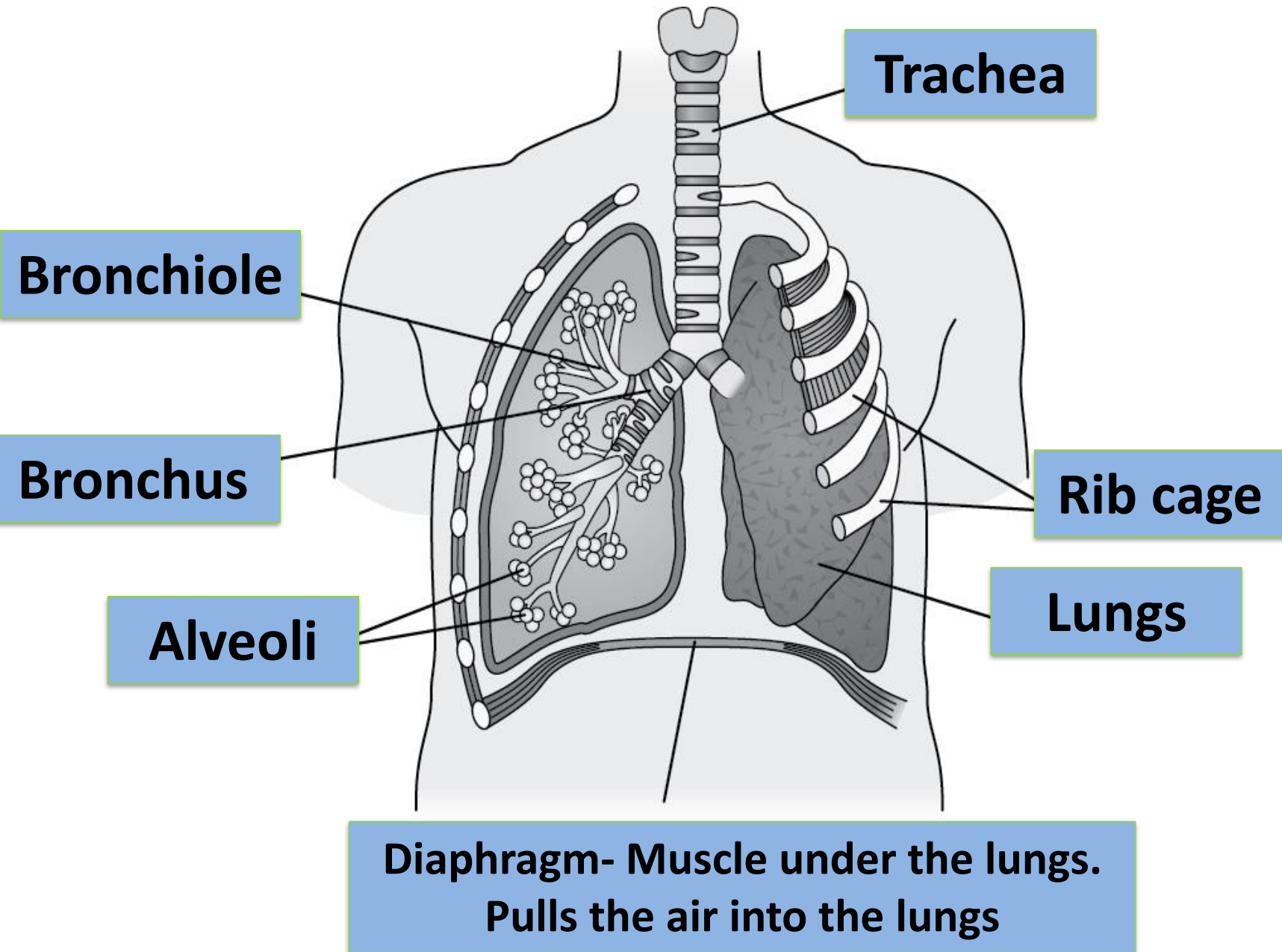
Alveoli

**Diaphragm- Muscle under the lungs.
Pulls the air into the lungs**

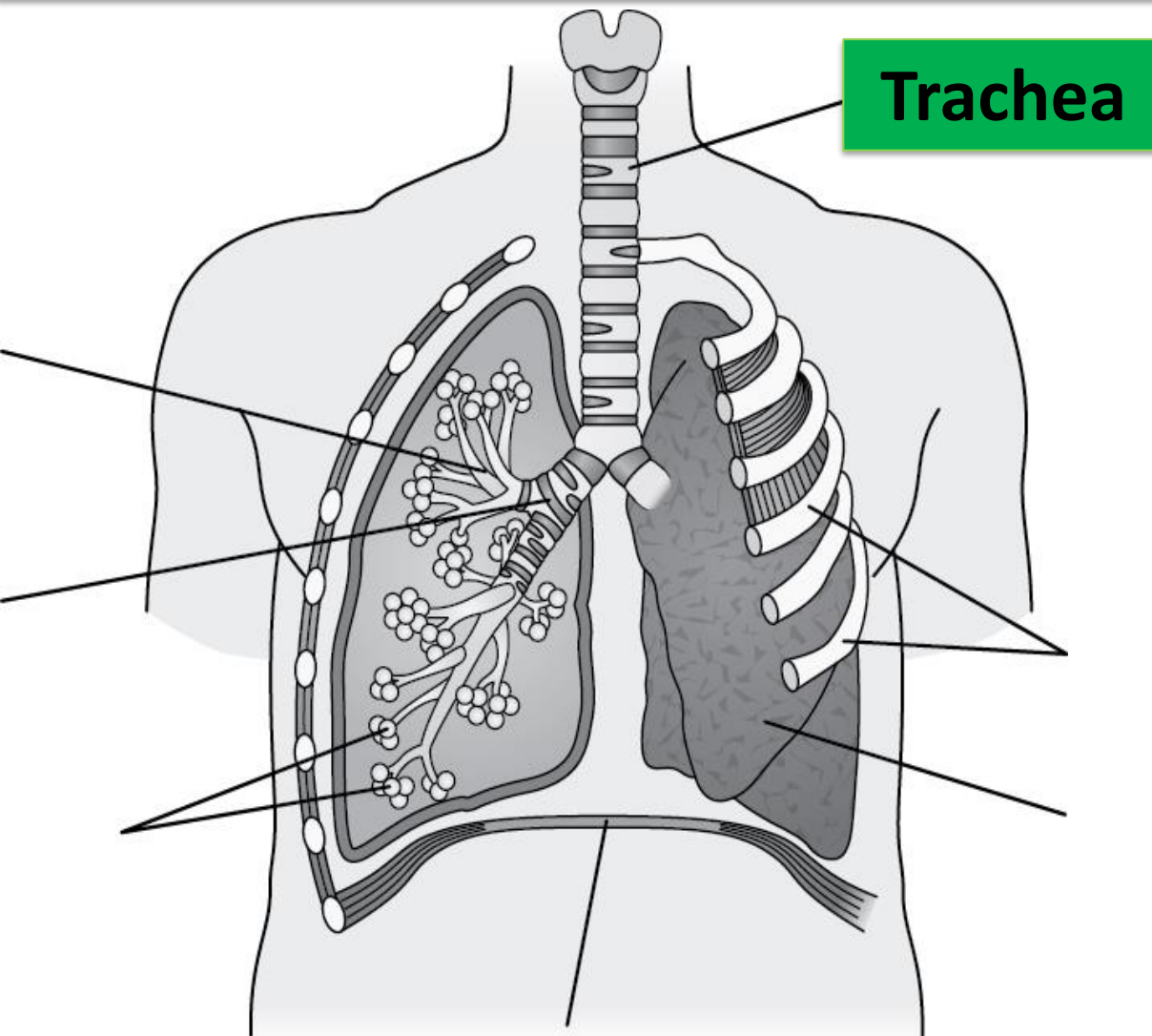
Rib cage



Review: The parts of our respiratory system



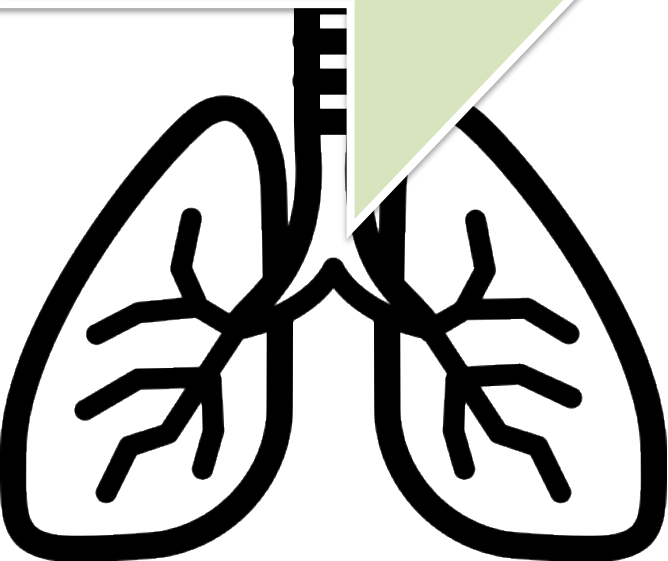
The **trachea** has **rings of cartilage** to **stop** the windpipe from **collapsing** under the **pressure** of breathing.



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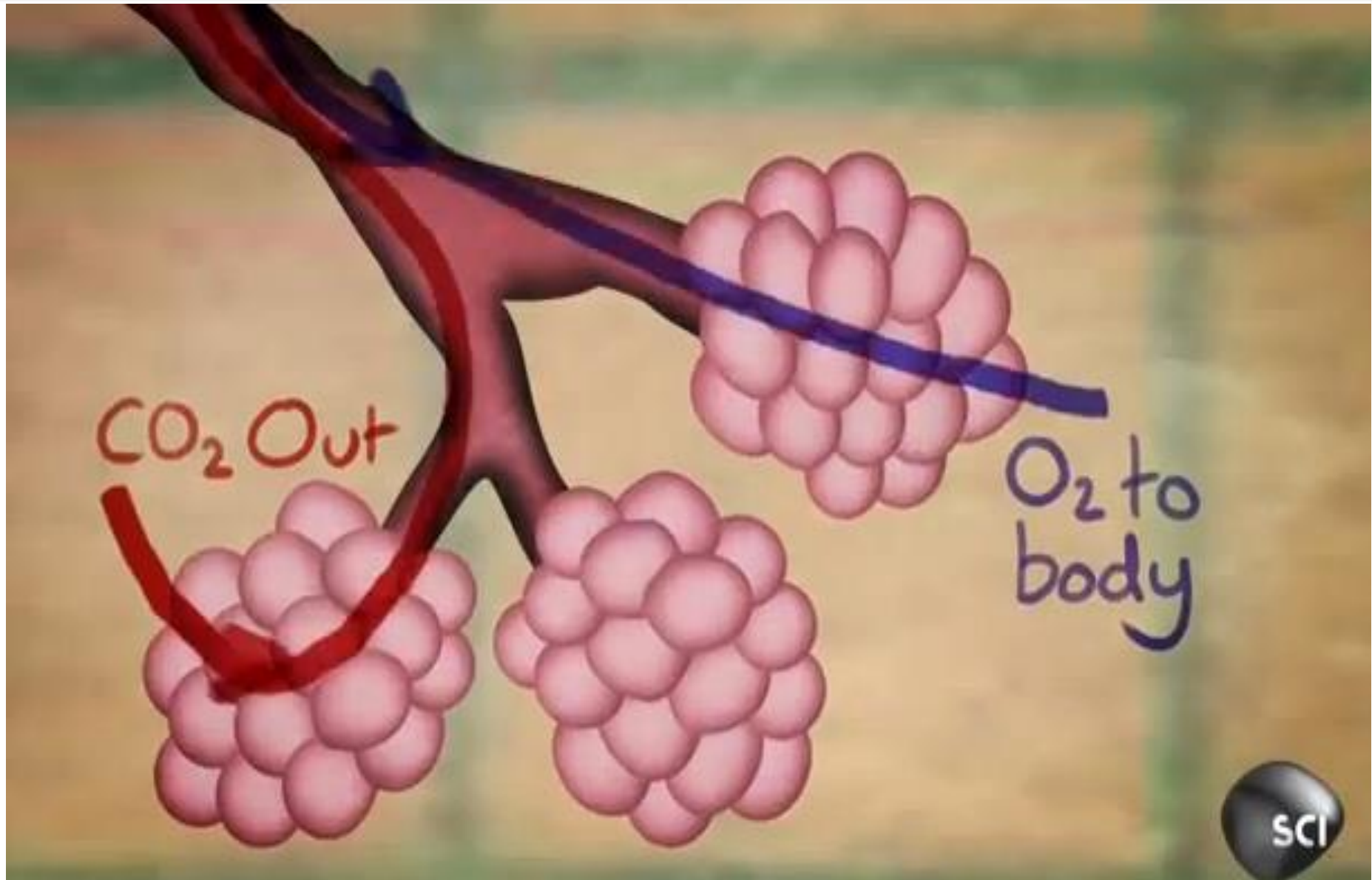


Learning Outcomes:

- ✓ Recall what gas exchange is and why we continually breathe
- ✓ Describe the structure of the gas exchange system
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Did you know... the lungs are not the same size. The right lung is larger than the left lung to help the heart fit into the rib cage.

The alveoli: where gas exchange happens.



Activity: As you watch the video complete your sheet

WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

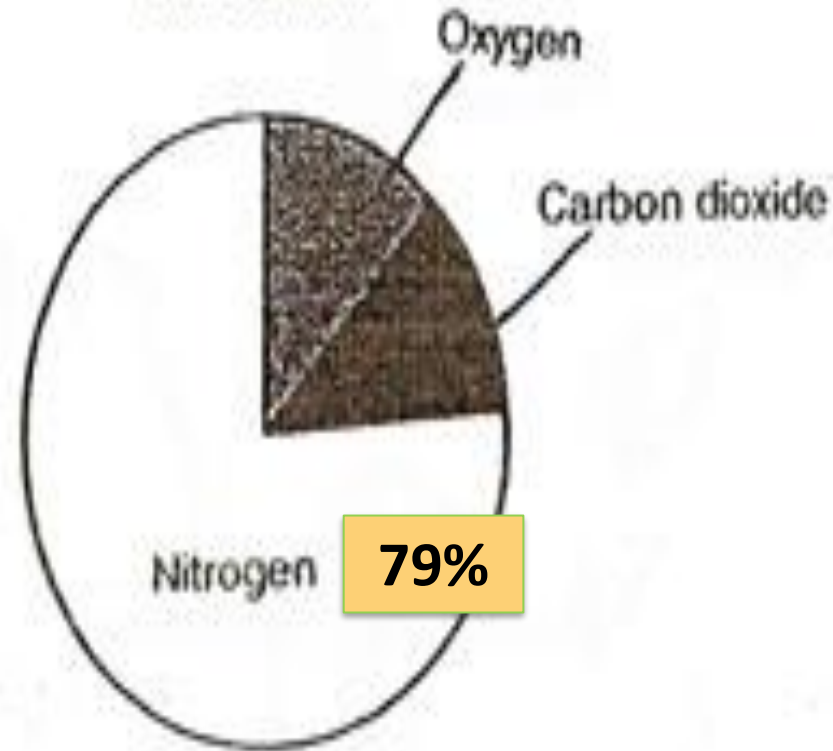
POWER!

Review

- ☐ **Large surface area** – to increase the volume of gas exchanged
- ☐ **Very thin** – One cell thick. Faster diffusion as it is a shorter distance for gas to travel
- ☐ **Rich supply of capillaries** – maintains the highest concentration gradient possible
- ☐ **Moist lining** – faster diffusion

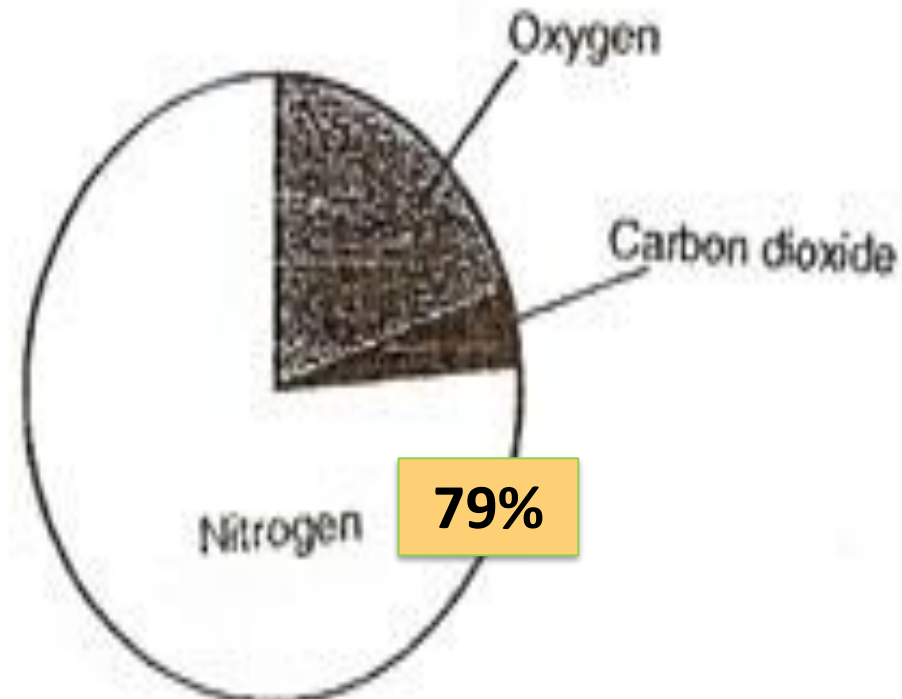
Which pie chart shows the amount of different gases found in our inhaled air?

Pie Chart A



Exhaled

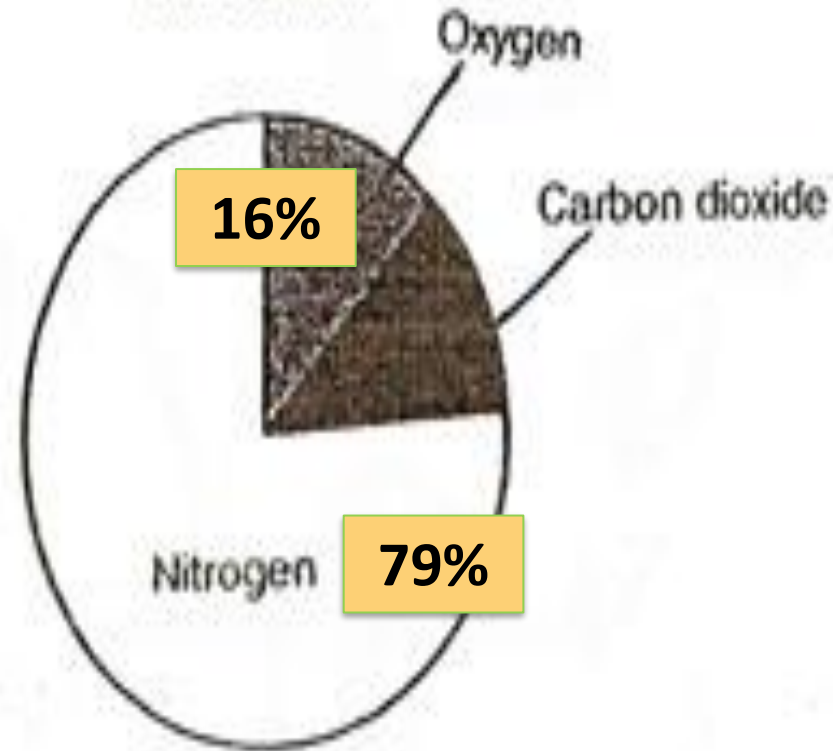
Pie Chart B



Inhaled

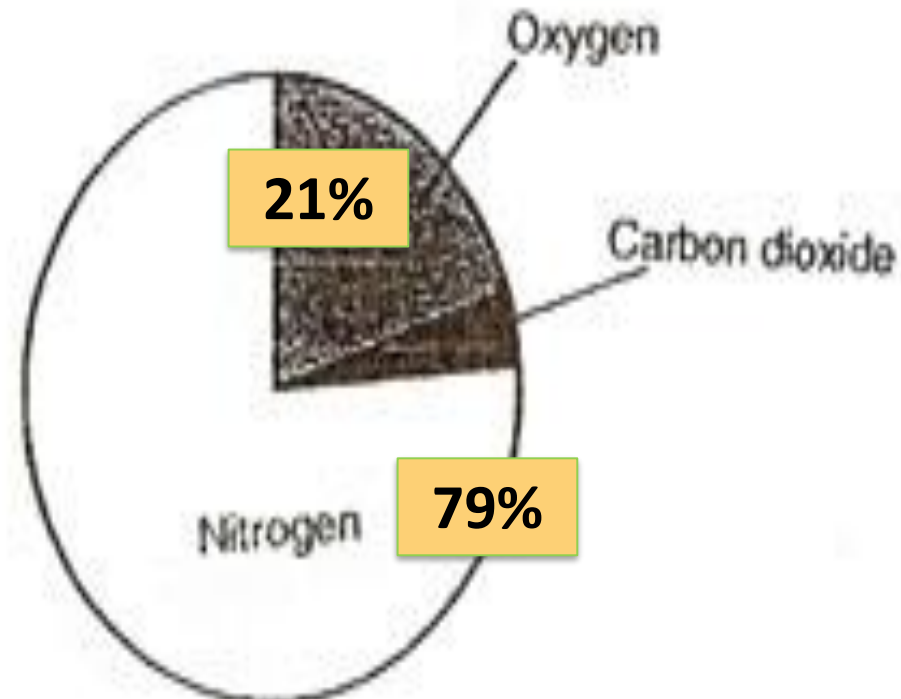
Oxygen is less in our exhaled air because some of it is used by our body by **respiration**. Which is a process that **transfers energy from food**.

Pie Chart A



Exhaled

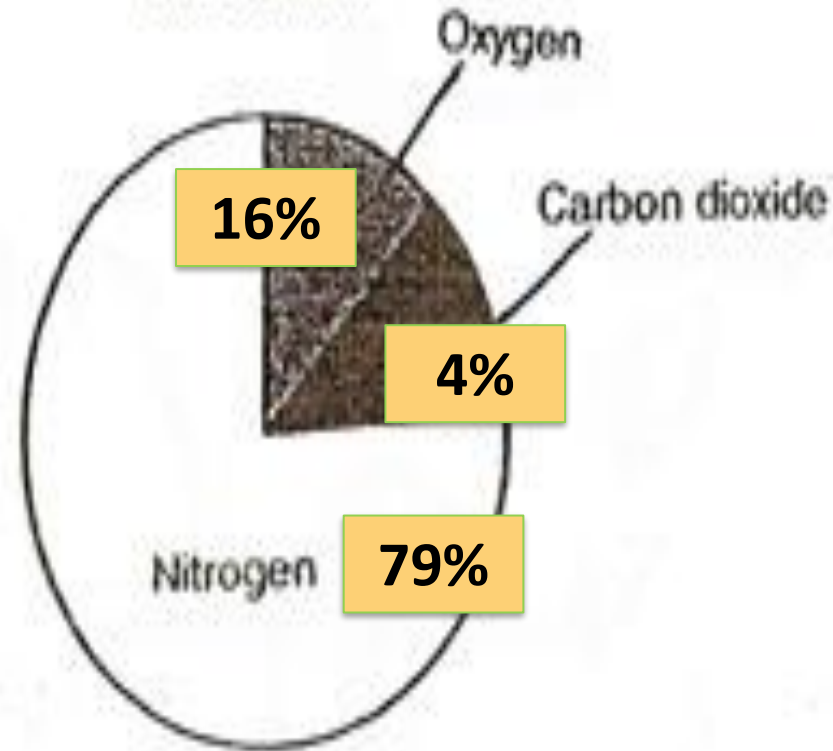
Pie Chart B



Inhaled

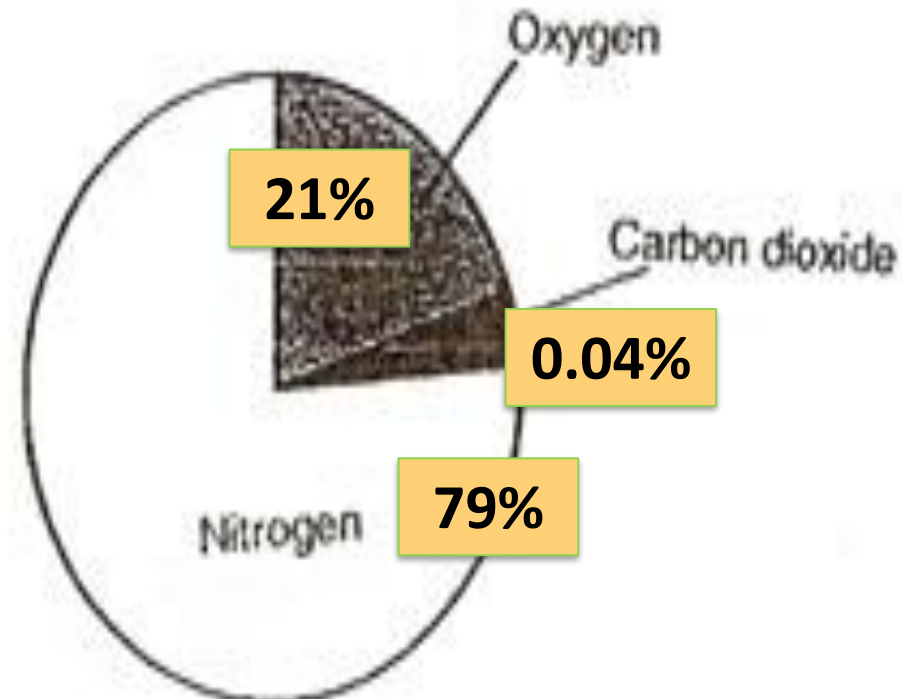
Respiration makes carbon dioxide as a waste product which needs to be removed in our exhaled air.

Pie Chart A



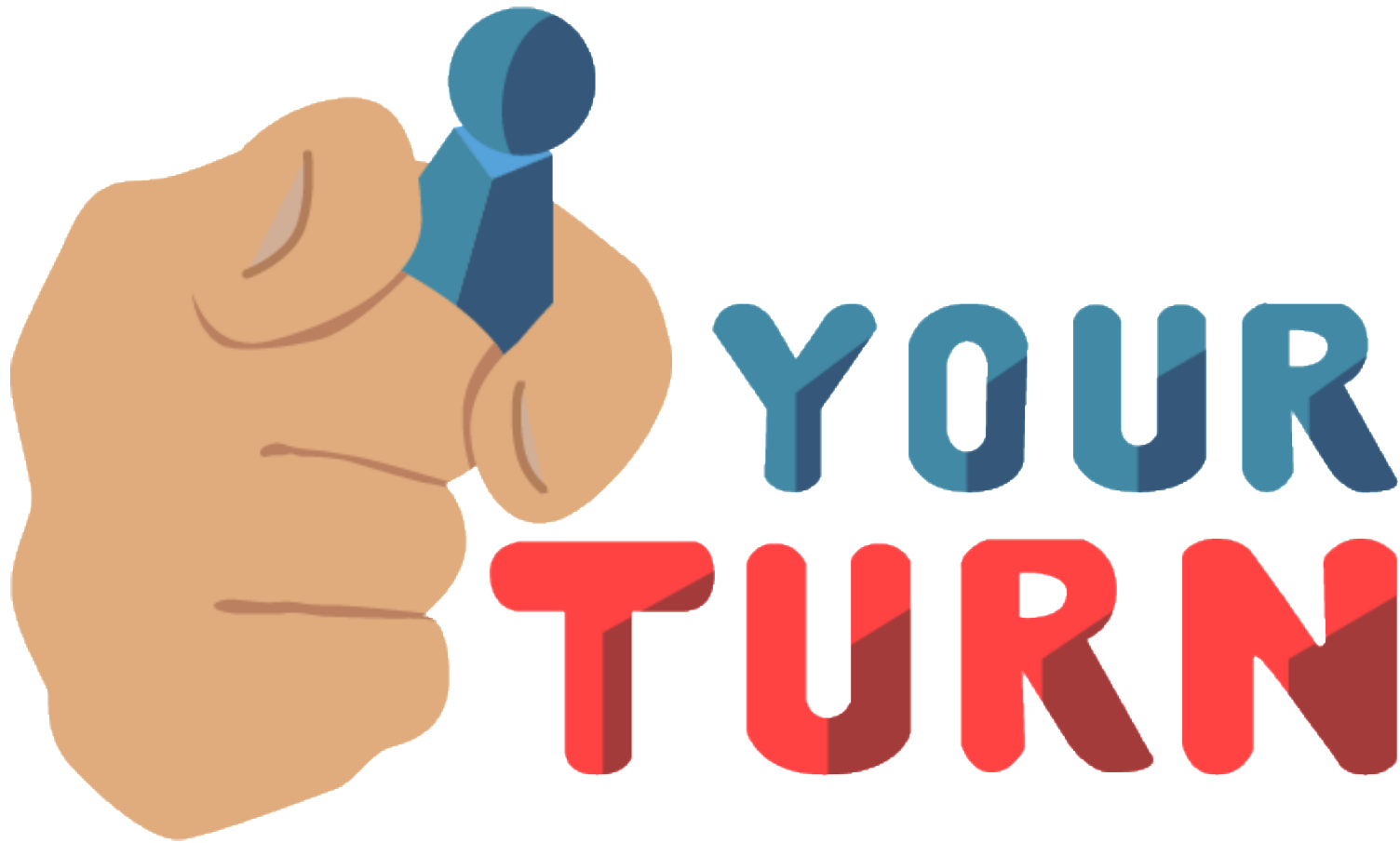
Exhaled

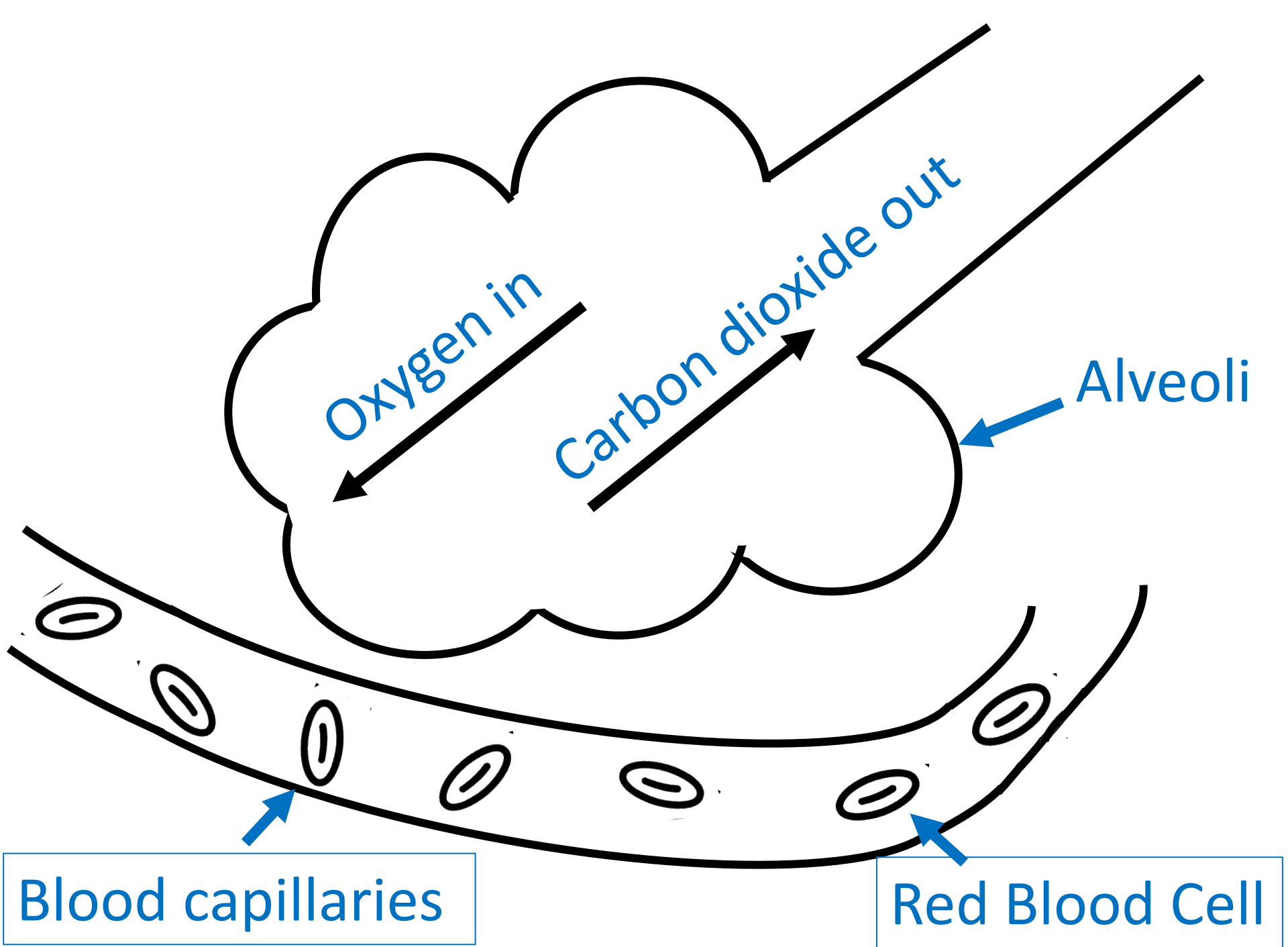
Pie Chart B

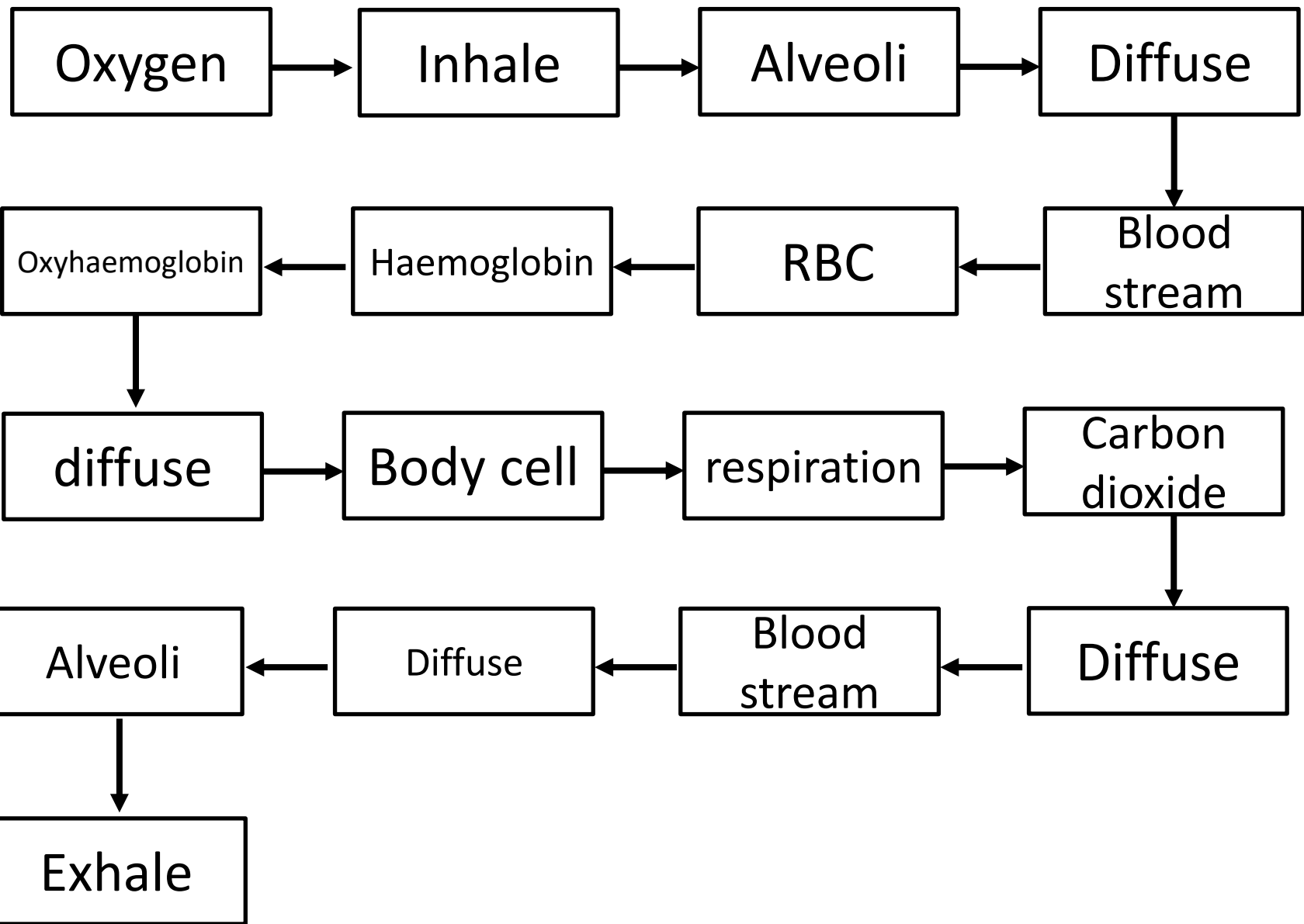


Inhaled

Activity: Complete your worksheet





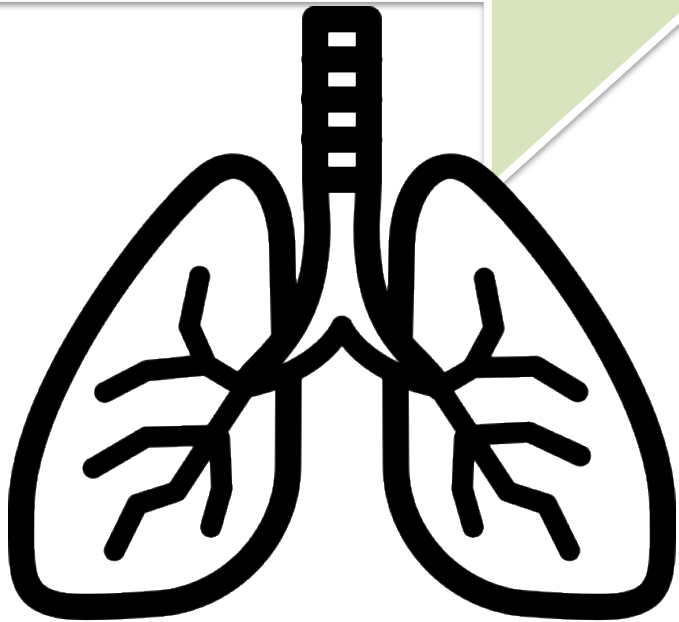


Review

Structure	Inhaling	Exhaling
Diaphragm	Contracts downwards	Relaxes up
Chest volume	Increased	decreased
Rib cage	Pull up and out	Pull down and in

LOb: Understand the respiratory system and how it is involved in gas exchange.

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Learning Outcomes:

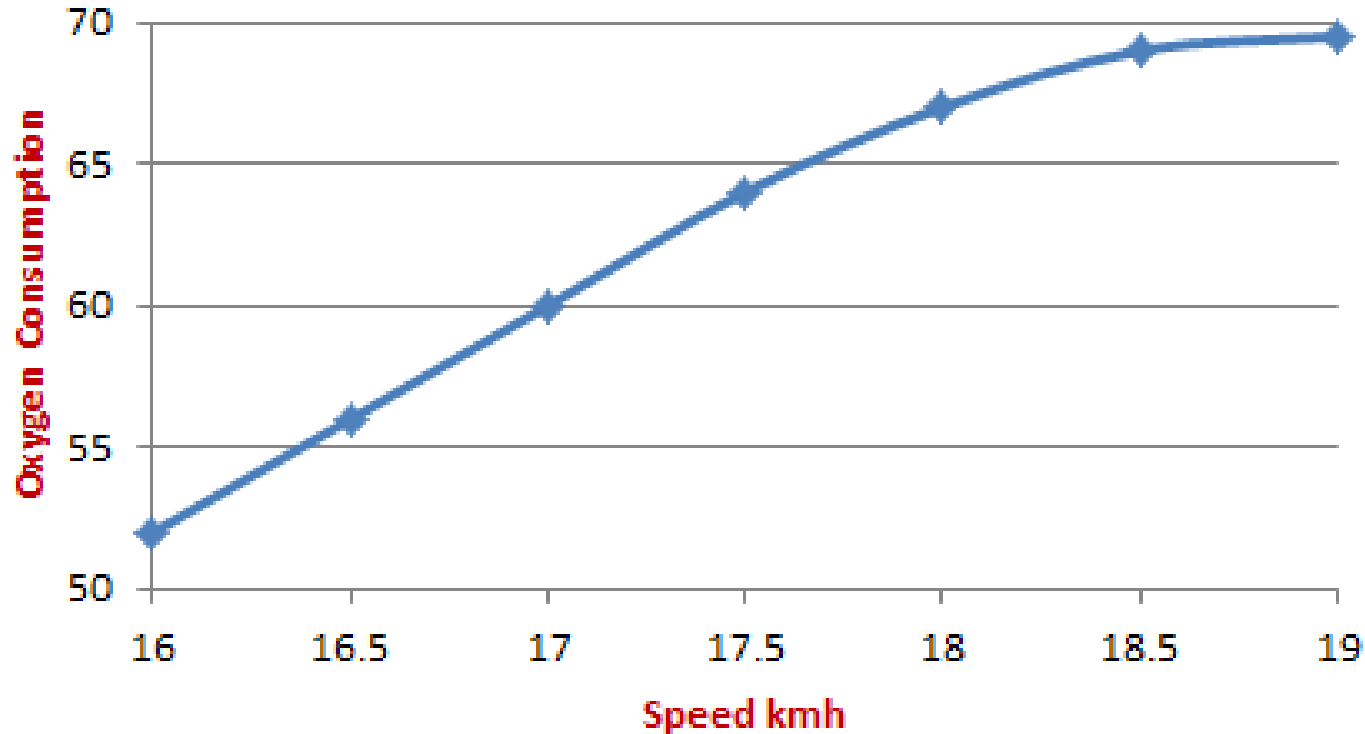
- ✓ Recall what gas exchanges is
- ✓ Describe the structure and adaptations of the gas exchange system
- ✓ Explain why we inhale and exhale

Did you know... the lungs are not the same size. The right lung is larger than the left lung to help the heart fit into the rib cage.

Exam question: Data was collected for oxygen consumption during exercise. Plot a graph of this information and then describe the relationship.

Speed (kph)	Oxygen consumed (L/min)
16	52
16.5	56
17	60
17.5	64
18	67
18.5	69
19	69

Speed vs Oxygen Consumption



As the speed increases the increase in oxygen consumption is directly proportional up to 17.5 kmh. After this speed the increase in oxygen consumption decreases until it levels off (plateaus).

keyfacts



Gas exchange is when we **take in oxygen** and **breathe out carbon dioxide**.



Air's final destination is the **alveoli** in the **lungs** which then diffuses into the **red blood cells in the blood stream**.



Alveoli adaptations = **large surface area**, very **thin** and having a **moist lining and good capillary blood supply**.



Continued breathing maintains a high concentration Gradient for maximum efficiency.

Plenary: People often say we breathe in oxygen and breathe out carbon dioxide explain why they are wrong.

Atmospheric gas	% gas inhaled	% gas exhaled
Nitrogen	80	80
Oxygen	21	16
Carbon dioxide	0.04	4



Plenary: Choose two boxes

1. The most difficult
part of today's lesson
was ...

10. I think I am
becoming an expert
on this topic
because ...

12. I was surprised to
learn that ...